Operations Strategy:

Among the most important decisions you will make.

In-House? **Outsource?** **In-License?** **Out-License?**

Penn Engineering

EAS5450: Class #16

1

Operations Strategy: *there may not be a single "right" answer, but there will be "wrong" ones…*

Add value by:
- Leveraging our core competencies
- Mitigating our weaknesses

Choose a strategy that:
- Provides for market entry, and
- Creates a defensible position once there

Update the operations strategy:
- As our capabilities change
- As our markets change

2

Recall…

Operations Strategies

Integrator

Description
Manage all the steps necessary to generate profits from an idea.

Investment requirements
High. Capital may be needed to set up new manufacturing facilities, for instance.

Capability requirements
- Strong cross-functional links within organization
- Product design
- Manufacturing-process design skills
- Technical talent sourcing

Best used when
- speed-to-market is not critical.
- technology is proven.
- customer tastes are stable.
- innovation is incremental.

Orchestrator

Description
Focus on some steps and link with partners to carry out the rest.

Investment requirements
Medium. Capital may be needed only to market the product, for example.

Capability requirements
- Ability to collaborate with several partners simultaneously, while not having direct control
- Complex project-management skills
- Customer insight
- Brand management
- Culture that can let go of certain areas, while focusing on core competencies
- Ability to move quickly; nimbleness

Best used when
- there is a mature supplier/partner base.
- there is intense competition – a need for constant innovation.
- strong substitutes exist.
- technology is in early stages.

Licensor

Description
License the innovation to another company to take it to market.

Investment requirements
Low. Manufacturing and marketing expenses are borne by other companies.

Capability requirements
- Intellectual-property management skills
- Basic research capabilities
- Contracting skills
- Ability to influence standards

Best used when
- there is strong intellectual property protection.
- importance of innovator's brand is low.
- market is new to the innovator.
- significant infrastructure is needed but not yet developed.

Andrew, James P. and Harold L. Sirkin. "Innovating for Cash." *Harvard Business Review.* Sept 2003

3

Make vs. Buy

Perform a task in-house if the answers to most of these questions is yes…

- Does this task involve proprietary or special skills unique or special to you?
- Does this task help to differentiate your product or service from the competition?
- Is this task critical to the perceived value of your product or service?
- Can you justify the fixed costs of property, plant and equipment required to keep this step in-house compared to the variable costs of out-sourcing it?
- Are there large transaction costs (money, time, inconvenience, etc.) involved with subcontracting this work?

4

Make vs. Buy

Outsource a task if the answers to most of these questions is yes…

- Do outside vendors have proprietary or special skills that involve this task?
- Is the task routine and/or mundane that does not differentiate your product?
- Does the task add little perceived value to your product?
- Can you purchase the task less expensively than you can do it yourselves?
- Can you justify the variable costs of outsourcing the task compared to the fixed costs of doing it in-house?

5

Operations Strategy:

Strategic Considerations

6

Considerations...

- Our Core Competency
  - Existing strengths (value added)
  - Organizational change
    - Culture clashes
  - Barriers to entry
- Our Growth Objectives
  - Acquiring new capabilities
  - Time horizons

7

Considerations (cont'd)…

- Strategic Alliances
  - Complementary resources
  - Spreads risk
  - Control issues
- Intellectual Property
  - I.P. risk
  - I.P. management (retain certain rights)
  - Technology transfer: to/from strategic partner

8

Considerations (cont'd)…

- Financial
  - Investment requirements: amount and ROI
  - Available resources
  - Risk tolerance
  - Preserve or give up the upside
  - Conserve, focus $ on core competency
- Timing
  - Does deferring a decision reduce risk?
  - Accelerating the path to market

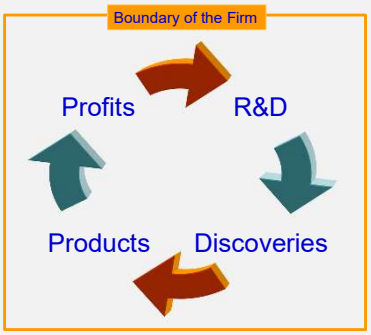
9

Considerations (cont'd)…

- Management Diversion
  - Operations management
  - Control
  - Financial management
  - Human resource management
- Morale

10

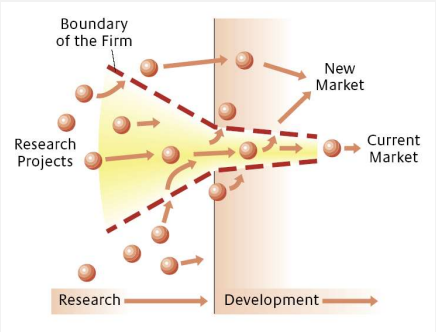
Closed Innovation Model



Henry Chesbrough, "The Era of Open Innovation." MIT Sloan Management Review. Spring 2003.

11

Open Innovation Model



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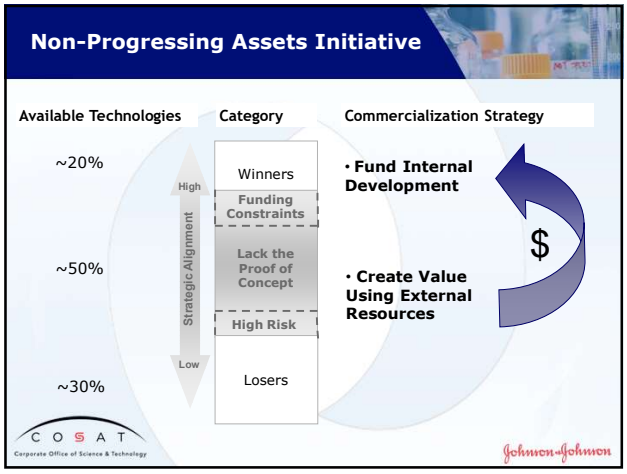
The Era of Open Innovation

- **Closed Model**
  - Self-reliant
  - Controls I.P.
  - Weeds out *false positives*, but…
  - Fails to capture *false negatives*
  - Myopic and insular
- **Open Model**
  - Strategic alliances
  - Licensing
  - Weeds out *false positives*, and…
  - Captures the value of *false negatives*
  - Entrepreneurial opportunity

13



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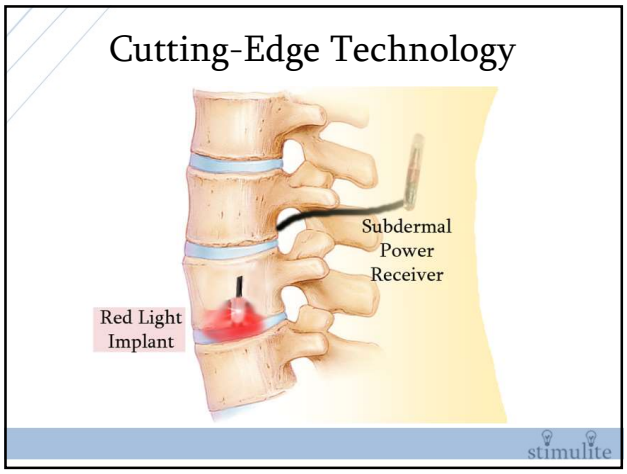

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J&J Buys Itself a Pipeline

It's snapping up biotechs, spearheading innovative alliances, and reshuffling R&D.

A year ago, J&J was losing patent protection on two giant products: treatments for epilepsy and schizophrenia. It didn't have a single product on the near horizon that could make up for the $7 billion in annual sales that would disappear when the two drugs went generic.

Sheri McCoy, head of J&J's pharmaceutical division, poured her energies into acquisitions, partnerships, and novel research collaborations, hoping to gain faster access to promising new markets. It closed a $1 billion deal for Cougar Biotechnology in July, so it now owns the up-start's promising array of cancer medications. In early September it spent $885 million for an 18% stake in Elan Pharmaceuticals. And J&J will pay $441 million for 18% of Crucell, a Dutch biotech company that's developing vaccines. These deals follow an ongoing string of successful acquisitions starting a decade ago. Analysts expect J&J to continue making modest acquisitions of promising smaller companies while shunning the megadeals that some of its rivals have done.

BUSINESSWEEK: October 12, 2009

16

FierceBiotech
THE BIOTECH INDUSTRY'S DAILY MONITOR

Published on FierceBiotech (http://www.fiercebiotech.com)

Merck beefs up bankroll for biotech startups

June 4, 2012 | By Ryan Mc

Merck has emerged as one
stage venture funds, and th
woes of newly hatched bio
($MRK) could boost resea
have abandoned for safer b

As MIT's *Technology Revi*
backed a $270 million fund
companies, dipping into a
gamble on research at your
of Merck's venture money
R&D programs. It could pe
as startups try to develop n

Drugmakers, of course, hav
some companies seeking e
are taking a more hands-of
as part of broader strategy.

FierceBiotech
THE BIOTECH INDUSTRY'S DAILY MONITOR

Published on FierceBiotech (http://www.fiercebiotech.com)

Pfizer ramps up work with academics and biotech startups

June 15, 2012 | By Ryan McBrid

Now more than a year into an R&
Pfizer ($PFE) has won over some
advancing science from labs to th
Harvard scientist Hal Dvorak tra
Pfizer, which has become entrenc
San Francisco and San Diego.

Pfizer's Centers for Therapeutic I
pharma giant to work with acade
previous arrangements with univ
projects. Essentially, Pfizer awar
gain proof-of-mechanism data fo
an option to develop the asset.

For Pfizer, the center speaks to a
drug development more efficient
giants such as Sanofi ($SNY), M

FierceBiotech
THE BIOTECH INDUSTRY'S DAILY MONITOR

Published on FierceBiotech (http://www.fiercebiotech.com)

GlaxoSmithKline mounts an open casting call for R&D starlets

March 25, 2015 | By Damian Garde

GlaxoSmithKline ($GSK) is looking for some new partners in academia, launching a contest for would-be collaborators with early-stage projects that could translate into marketable medicines.

The U.K. drugmaker is now accepting proposals for its third annual Discovery Fast Track Challenge, asking scientists in Europe, Canada and the U.S. to send in pitches for how their discovery-stage idea could blossom into a clinical program. GSK plans to pick as many as 12 winners from the pool, inviting their inventors to screen each target against the company's extensive compound library. If there's a hit, GSK could offer winners a collaboration deal through its Discovery Partnerships with Academia (DPAc) program, moving forward together after crossing out terms.

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FierceBiotech
THE BIOTECH INDUSTRY'S DAILY MONITOR

Targeting ALS, GlaxoSmithKline and Avalon seed another build-to-buy startup

November 9, 2015 | By Damian Garde

Partners GlaxoSmithKline ($GSK) and Avalon Ventures have seeded a 7th startup in their company-building collaboration, lining up $10 million for a new effort to treat ALS.

Avalon Ventures Managing Director Jay Lichter

The company, Iron Horse Therapeutics, is taking a novel approach to the debilitating disease with technology licensed from Sanford Burnham. Patients with ALS, or amyotrophic lateral sclerosis, suffer a gradual weakening of muscles that eventually proves fatal. But those that have a gene defect that limits production of the protein EphA4 tend to live longer, according to the company, and Iron Horse is at work developing small-molecule therapies that could block EphA4 to lessen the impact of the disease.

Now, thanks to GSK and Avalon, the company is in line for up to $10 million in Series A funding as it works through the earliest stages of development, with R&D help from both partners and the Avalon-run incubator COI Pharmaceuticals.

Like its 6 predecessors, Iron Horse begins on a track that could end with a GSK buyout. Under a partnership formed in 2013, Avalon and GSK co-support startups until they've selected a lead drug candidate, at which point the U.K. pharma giant has an exclusive right to acquire each company. If GSK passes, Avalon hangs on to its ownership and gets to choose whether to press forward.

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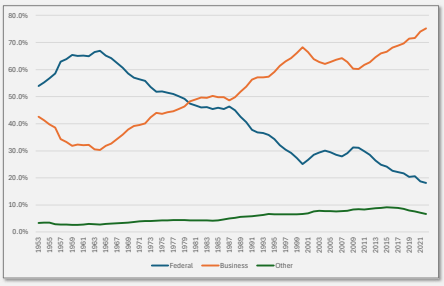
U.S. R&D Shift to Private Sector

Since 1980, more R&D funding has come from industry than government



Source: National Science Foundation, National Patterns of R&D Resources, Feb 2022

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Reshuffling U.S. Corporate R&D

Small companies are a growing source of R&D for large companies

Employees	1981	1989	1999	2018
< 1,000	**4.4%**	9.2%	22.5%	**19.0%**
1,000-4,999	6.1%	7.6%	13.6%	15.5%
5,000-9,999	5.8%	5.5%	9.0%	10.3%
10,000-24,999	13.1%	10.0%	13.6%	19.1%
> 25,000	**70.7%**	67.7%	41.3%	**36.0%**

Business R&D expenditures

Businesses with <100 employees accounted for 9.1% of all U.S. R&D spending in 2022

Source: NSF, Nat'l Center for Science & Engineering Statistics, 2022.

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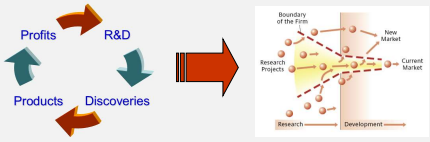
DRIVERS OF CHANGE



- Venture capital – corporate & independent
- Mobility & numbers of *knowledge workers*
- Mental & strategic shift to explore options

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Do it ourselves?

Outsource it to others?

Among the most important decisions you will make.

License it to others?

License it from others?

22

Palm Computing, Inc.

Financing Challenges

Jeff Hawkins

Donna Dubinsky

23

Palm Computing, Inc. Financing Challenges Chronology

	Event	Amount Raised	Post-Money Valuation	% of Company Sold	% of Company Owned by Mgmt
Nov 91	"Palm (A)" case: Tandy cross-license agreement				
Jan 92	1st round financing: Palm Computing launched.	$1,300,000	$3,250,000	40%	60%
Sept 92	2nd round financing	$3,000,000	$15,000,000	20%	48%
Late 93	3rd round financing commitment				
Aug 93	Apple Newton launched				
Oct 93	Tandy Zoomer launched (10,000 sold)				
Jan 94	3rd round financing	$3,000,000	$30,000,000	10%	43.2%
---	$1B lost in hand-held products industry-wide				
Aug 95	Today's "Palm Financing" case				
Feb 97	"Palm Pilot" case: customer-driven product				

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Palm Computing, Inc. at time of this case

- Visionary, passionate, technology leadership: Hawkins
- Management talent: Dubinsky
- Patented technology
- Integrated business model
- Fully-developed whole product
  - 5 months from launch
- Tandy relationship
- VC backing
- $1.5MM on hand
  - Low burn rate: ~$250K/mo
- Founders still own 43.2%

25

Net Margin to Palm Computing…

Retail Price Point	$300.00
30% Retail Gross Margin	- 90.00
Wholesale Price	210.00
5% Commissions & Discounts	- 10.50
Gross Margin to Palm	199.50
C.O.G.S. + 5% Allowance	-105.00
5% Commissions & Discounts	- 10.50
Gross Margin to Palm	199.50
C.O.G.S. + 5% Allowance	-105.00
Net Margin to Palm	$ 94.50 per unit sold

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Palm Computing: 9-Month Cash Flow Projection

	Aug-95	Sep-95	Oct-95	Nov-95	Dec-95	Jan-96	Feb-96	Mar-96	Apr-96
						Product Launch			
Beginning cash on hand	$1,500,000	$1,250,000	$1,000,000	$450,000	($250,000)	($1,250,000)	($2,700,000)	($3,800,000)	($3,955,000)
Unit sales						*10,000*	*15,000*	*20,000*	*25,000*
Net Margin to Palm						*$94.50*	*$94.50*	*$94.50*	*$94.50*
Revenues								945,000	1,417,500
Burn rate	(250,000)	(250,000)	(300,000)	(350,000)	(400,000)	(450,000)	(500,000)	(500,000)	(500,000)
Tooling			(250,000)	(250,000)					
Marketing				(100,000)	(100,000)	(500,000)	(100,000)	(100,000)	(100,000)
Inventory					(500,000)	(500,000)	(500,000)	(500,000)	(500,000)
Ending cash on hand	$1,250,000	$1,000,000	$450,000	($250,000)	($1,250,000)	($2,700,000)	($3,800,000)	($3,955,000)	($3,637,500)

Notes:
- A. Unit sales begin at 10,000/mo and grow 5,000/mo (based on Zoomer launch).
- B. Revenues based on $94.50/unit average net margin. Retail channels pay in 60-90 days.
- C. Burn rate increases $50,000/mo beginning in Oct for pre-launch ramp-up (sales support, tech support, repair, etc.). Levels off after Jan. launch.
- D. Tooling ordered in Nov (50% paid with order & 50% in 30 days).
- E. Marketing begins in Nov @ $100,000/mo pre-launch for brochures, PR, sales tours, user manuals, packaging, etc. Increases to $500,000 at Jan. launch. Thereafter, levels off at $100,000/mo.
- F. Inventory build of 5,000 units/mo @ $100 cost/unit.

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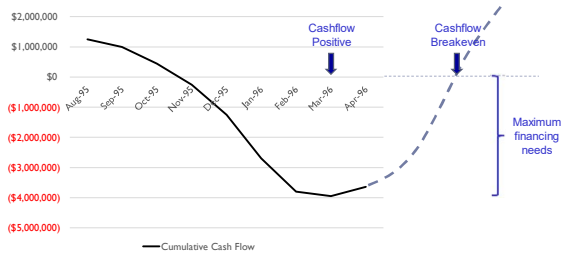
Cumulative Cash Flow Determines Funding Needs



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	Valuation	Hawkins & Mgmt	Merrill-Pickard	Sutter Hill	Tandy	Newtek Ventures	Credit Lyonnais	ABC Venture
ROUND 1 (JAN 92)								
New investment:	$1,300,000		$500,000	$500,000	$300,000			
% Ownership:	40.0%	60.0%	15.4%	15.4%	9.2%			
Value:	$3,250,000	$1,950,000	$500,000	$500,000	$300,000			

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	Valuation	Hawkins & Mgmt	Merrill-Pickard	Sutter Hill	Tandy	Newtek Ventures	Credit Lyonnais	ABC Venture
ROUND 1 (JAN 92)								
New investment:	$1,300,000		$500,000	$500,000	$300,000			
% Ownership:	40.0%	60.0%	15.4%	15.4%	9.2%			
Value:	$3,250,000	$1,950,000	$500,000	$500,000	$300,000			
ROUND 2 (SEP 92)								
New investment:	$3,000,000		$750,000	$750,000		$1,500,000		
Total investment:			$1,250,000	$1,250,000	$300,000	$1,500,000		
% Ownership:	20.0%	48.0%	17.3%	17.3%	7.4%	10.0%		
Value:	$15,000,000	$7,200,000	$2,596,154	$2,596,154	$1,107,692	$1,500,000		

30

	Valuation	Hawkins & Mgmt	Merrill-Pickard	Sutter Hill	Tandy	Newtek Ventures	Credit Lyonnais	ABC Venture
ROUND 1 (JAN 92)								
New investment:	$1,300,000		$500,000	$500,000	$300,000			
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% Ownership:	20.0%	48.0%	17.3%	17.3%	7.4%	10.0%		
Value:	$15,000,000	$7,200,000	$2,596,154	$2,596,154	$1,107,692	$1,500,000		
ROUND 3 (JAN 94)								
New investment:	$3,000,000						$3,000,000	
Total investment:			$1,250,000	$1,250,000	$300,000	$1,500,000	$3,000,000	
% Ownership:	10.0%	43.2%	15.6%	15.6%	6.6%	9.0%	10.0%	
Value:	$30,000,000	$12,960,000	$4,673,077	$4,673,077	$1,993,846	$2,700,000	$3,000,000	

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	Valuation	Hawkins & Mgmt	Merrill-Pickard	Sutter Hill	Tandy	Newtek Ventures	Credit Lyonnais	ABC Venture
ROUND 1 (JAN 92)								
New investment:	$1,300,000		$500,000	$500,000	$300,000			
% Ownership:	40.0%	60.0%	15.4%	15.4%	9.2%			
Value:	$3,250,000	$1,950,000	$500,000	$500,000	$300,000			
ROUND 2 (SEP 92)								
New investment:	$3,000,000		$750,000	$750,000		$1,500,000		
Total investment:			$1,250,000	$1,250,000	$300,000	$1,500,000		
% Ownership:	20.0%	48.0%	17.3%	17.3%	7.4%	10.0%		
Value:	$15,000,000	$7,200,000	$2,596,154	$2,596,154	$1,107,692	$1,500,000		
ROUND 3 (JAN 94)								
New investment:	$3,000,000						$3,000,000	
Total investment:			$1,250,000	$1,250,000	$300,000	$1,500,000	$3,000,000	
% Ownership:	10.0%	43.2%	15.6%	15.6%	6.6%	9.0%	10.0%	
Value:	$30,000,000	$12,960,000	$4,673,077	$4,673,077	$1,993,846	$2,700,000	$3,000,000	

USR PROPOSAL	$50,000,000	$21,600,000	$7,788,462	$7,788,462	$3,323,077	$4,500,000	$5,000,000	
ROUND 4 (AUG 95)								
New investment:	$4,000,000		$1,000,000	$1,000,000		$1,000,000		$1,000,000
Total investment:			$2,250,000	$2,250,000	$300,000	$2,500,000	$3,000,000	$1,000,000
% Ownership:	44.4%	24.0%	19.8%	19.8%	3.7%	16.1%	5.6%	11.1%
Value:	$9,000,000	$2,160,000	$1,778,846	$1,778,846	$332,308	$1,450,000	$500,000	$1,000,000

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Palm Computing, Inc. - What If ?

Annual sales of the Palm Pilot		750,000
Average retail price (mix of $299 and $399 units)	$325.00	
Retailers' gross margin (@ 30%)	-97.50	
Wholesale price	227.50	
Cost of goods sold per unit	-100.00	
Net margin per unit	$127.50	
Annual net margin		$95,625,000
Overhead and marketing		-$30,000,000
EBT (Earnings before taxes)		$65,625,000
PAT (Profit after taxes)		$42,656,250
PE Ratio (Price/Earnings) in Industry		30X
Potential IPO Valuation		$1,279,687,500

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Palm Computing, Inc. - Epilog

	Valuation	Hawkins & Mgmt	Merrill-Pickard	Sutter Hill	Tandy	Newtek Ventures	Credit Lyonnais	ABC Venture
USR PROPOSAL	$50,000,000	$21,600,000	$7,788,462	$7,788,462	$3,323,077	$4,500,000	$5,000,000	
ROUND 4 (AUG 95)								
New investment:	$4,000,000		$1,000,000	$1,000,000		$1,000,000		$1,000,000
Total investment:			$2,250,000	$2,250,000	$300,000	$2,500,000	$3,000,000	$1,000,000
% Ownership:	44.4%	24.0%	19.8%	19.8%	3.7%	16.1%	5.6%	11.1%
Value:	$9,000,000	$2,160,000	$1,778,846	$1,778,846	$332,308	$1,450,000	$500,000	$1,000,000
IPO (1996)								
Value:	$1.28 Billion	$307,200,000	$252,991,453	$252,991,453	$47,261,538	$206,222,222	$71,111,111	$142,222,222

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Palm Computing, Inc. - Epilog

Sep-95	U.S. Robotics acquires Palm Computing, Inc.
1996	Palm sells 350,000 Pilot Organizers
Mar-97	3-Com acquires U.S. Robotics for modem technology
May-98	Hawkins and Dubinsky leave Palm
Jul-98	Hawkins and Dubinsky form Handspring
Oct-98	Handspring Series A: $18,000,000 (Benchmark & Kleiner Perkins)
Jul-99	Handspring Series B: $10,000,000
Oct-99	Handspring launches Visor
Mar-00	3-Com spins off Palm with IPO @ $38/shr
Jun-00	Handspring IPO @ $20/shr
Jun-03	Palm @ $14.44/shr & Handspring @ $1.28/shr: merger announced
Oct-03	Merged company trades on NASDAQ

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TECH

H-P Gambles on Ailing Palm

By Justin Scheck And Yukari Iwatani Kane

Updated April 29, 2010 12:01 a.m. ET

Hewlett-Packard Co. HPQ -1.25% scooped up Palm Inc. for about $1 billion in cash, pushing the computer giant deeper into the competitive smartphone market and ending the independence of a struggling company that was rapidly running out of prospects.

- 2011: HP announces new Palm product line
- Aug 2011 HP ends production and support of Palm devices

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Meet the new Palm: Hint, this isn't your old PalmPilot

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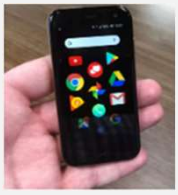

The new Palm, a small San Francisco startup that has licensed the name, is selling, a palm-sized $349 Android device that is exclusive to Verizon Wireless and requires a separate Verizon data plan and smartphone (an iPhone or Android). But it's not as straightforward as just another smartphone – it's not oversized like the top sellers, and it's not really even a smartphone.

The new Palm next to a

Instead, Palm is looking to define a new ultramobile market segment, which positions its tiny but cute titanium or gold, rounded-edge handset somewhere between an Apple Watch or some other connected wearable and one of the many megasized smartphones now in vogue.

"This is not a nostalgia play, this is a reinvention of the brand," insists new Palm co-founder Dennis Miloseski, whose resume includes stints at Google, Samsung and GE. But he also adds that, "we thought it was important to do justice to the Palm name."

https://www.palm.com/product

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